Supplementary Material S2

Machine-learning surrogates of MOF water-adsorption beds: the coefficient map, not the basis, limits accuracy on unseen materials

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This supplement holds supporting detail for the main text: each section expands a statement the main text summarises, with the same numbers. Section, figure, table and equation numbers without an "S" refer to the main text. Supplementary S1 is the correction ledger, whose entry codes (An, Bn) are cited below.

The terms are those of the main text (Section 1). The *legacy design* is the first dataset, with 12 held-out materials; the *smooth design* (v2) is the second, 240 materials in five folds, and the *MOF-like design* (v3) the third (Section 2.2). The rungs L1–L7 are the seven pre-registered candidate explanations of the error (Section 3.1). The *anchor* is the comparison of the reference solver with a measured MOF-303 bed (Section 2.1). The *edge rule* says that a tuning optimum found at the edge of its grid does not count until the grid is extended past it.

S1 Henry's law across the parameter space

S1.1 Henry's law holds everywhere; the region where it dominates does not

Across all 240 sampled materials, $n > 1$ strictly for every material, so the cooperative term vanishes faster than c at zero concentration and the Henry constant is finite and positive everywhere. But the Henry constant $K_H = q_{\max} f_H b_H$, the slope of equation (4) at the origin, spans 0.0322–2.52 $\mathrm{mol\,kg^{-1}}$ per $\mathrm{mol\,m^{-3}}$, 1.9 decades, median 0.359. The concentration at which the cooperative term reaches 1 % of loading (the top of the accessible Henry region) reaches $10^{-0.9}$ of the feed at best and a median of $10^{-2.0}$, but lies below 10^{-3} of the feed for 73 materials and below 10^{-6} for 34. For those the isotherm is effectively Sips-like at every concentration the column visits. Since concentration is proportional to relative humidity at fixed temperature, the near-linear uptake regime of a median material ends at $10^{-2.0}$ of the feed humidity. The design guarantees thermodynamic validity everywhere, not an experimentally accessible linear regime.

S2 Where real water-harvesting frameworks sit

The sampled space is meant to contain the adsorbents a water harvester would use, so we placed published ones in it (Figure S1), on the four axes that are properties of the framework rather than of the pellet and the packing: step position, capacity, heat of adsorption and the fraction of capacity taken below the step. The values come from the primary literature, MOF-303 [1, 2], MOF-801 [3], CAU-10-H [4, 5], aluminium fumarate [1, 6], MIL-160(Al) [7–9] and Co_2Cl_2(BTDD) [10]; each point spans the values its sources report, which differ in sample,

Table S1: Inputs of the six named frameworks, as the registered rules produce them (`results/dd_b2_frameworks.json` and `_v3.json`). Capacity, heat and step are the means of the stated values and are the same under both rules; n comes from the stated rise interval, or the design median where none is stated (MIL-160(Al), Co_2Cl_2(BTDD)); f_H is the smooth design's median under the v2 rule and the sources' below-step fraction under the v3 rule. For MOF-303 and aluminium fumarate the rise interval was read from a published figure, and for CAU-10-H it is the stated range of the jump, not a 10–90 % width (Supplementary S1, entry B88). Pellet size, density and bed porosity take each design's medians.

framework	q_{max} (mol kg^{-1})	$-\Delta H$ (kJ mol^{-1})	RH_{step}	n (v2)	n (v3)	f_H (v2)	f_H (v3)
MOF-303	25.6	52.0	0.135	5.4	5.4	0.14	0.17
MOF-801	20.0	60.0	0.070	6.3	6.3	0.14	0.07
CAU-10-H	18.3	56.5	0.205	30.0	30.0	0.14	0.03
aluminium fumarate	23.0	52.1	0.258	16.5	16.5	0.14	0.05
MIL-160(Al)	21.5	50.5	0.085	3.5	16.4	0.14	0.06
Co_2Cl_2(BTDD)	53.7	55.0	0.280	3.5	16.4	0.14	0.19

temperature and in how the step is defined, and the below-step fractions are mostly read from published figures and are rough. Of the 6 frameworks, 3 lie inside the smooth design's sampled ranges on these four axes (the step steepness, which also puts aluminium fumarate outside, is discussed below). The exceptions are at the edges, and one is far outside: MOF-801's step, at a relative humidity of about 0.07, sits just below the sampled floor of 0.08; one reported heat for CAU-10-H, -61 kJ mol^{-1}, lies just beyond the sampled range; and Co_2Cl_2(BTDD), at about 54 mol kg^{-1}, holds 1.8× the largest capacity we sampled in the smooth design, 30. The cooperativity exponent n, the steepness of the step and arguably the property a harvester designer cares about most, is not on these axes, because the sources report isotherms, not fitted exponents. The named-framework test (Section 3.5) reads it from the stated width of the rise by a registered rule, which puts aluminium fumarate and CAU-10-H far above the smooth design's range and MOF-801 ($n = 6.3$) just above it. This section places each framework as the range of values its sources report, so one extreme report can sit outside; Table 2 of the main text classifies by the rule's mean inputs instead, which is why it lists CAU-10-H as outside on n only and MOF-801 on its step, heat and n; the MOF-like design was built to enclose all six frameworks on every axis, n included. Table S1 lists the inputs both rules produce.

S3 The tuning envelope of rung L3

The tuning envelope is itself a result

The families optimise 3.0 orders of magnitude apart: the Chebyshev family at 0.03 and the perceptron at 3×10^{-5}. 3 configurations failed on every seed and 4 more on some seeds; all are excluded and listed, never averaged in. Every family, the perceptron included, was tuned over the same learning-rate grid; none was left at a default.

S4 Basis size on the legacy design

The legacy result below is retracted as a verdict (A29) and kept as a measurement. At matched parameter counts and each family at its own best learning rate over 8 rates spanning 2.8 decades, DeepONet looked flat in basis size (Figure S2): -0.00102 from $p = 8$ to $p = 128$, interval $[-0.00585, 0.00491]$, which spans zero. Stated as what it excludes: a sixteenfold increase in basis size could not have improved this arm by more than 19 % of its error. The smooth design (v2) found an improvement well inside that bound, which is exactly why a null on 12 held-out materials could not be the last word. Meanwhile its own POD floor falls from 1.44×10^{-2} to

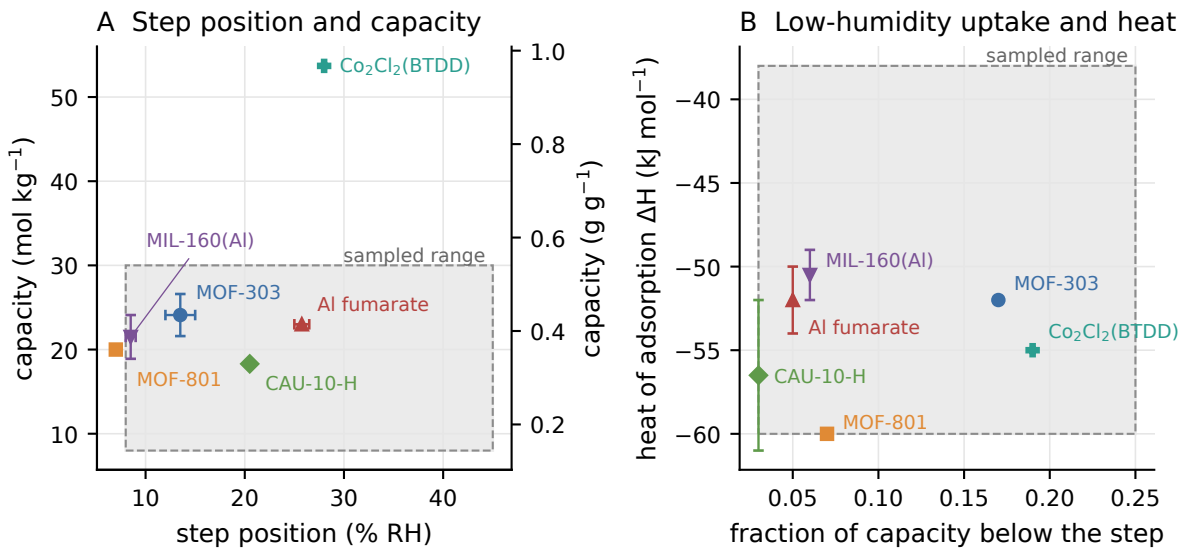


Figure S1: Published water-harvesting frameworks against the sampled parameter space (shaded boxes). **A** Step position and capacity. **B** Fraction of capacity taken below the step and heat of adsorption ΔH (negative: adsorption is exothermic). Bars span the values reported across sources; values, locations in each paper and confidence are recorded in `research/water_mof_parameters.json`.

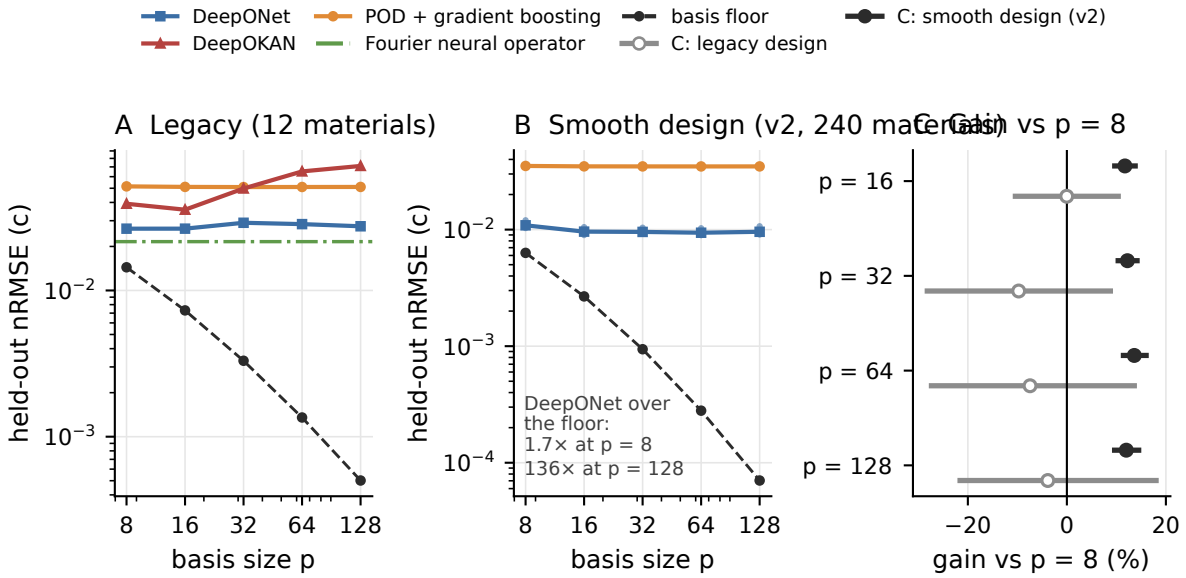


Figure S2: Basis size on both designs. **A** Legacy design (12 held-out materials): held-out error of each operator family against basis size p, with the basis floor; the Fourier operator has no basis size and is drawn as a level. **B** Full design (240 materials, five folds): DeepONet against p (one faint point per fold, line through the means), the total error of POD with per-mode gradient boosting (basis floor included; the map term alone is in the main text, Fig. 4), and the basis floor. The floor falls by orders of magnitude; the total error does not move, and DeepONet improves only from $p = 8$ to $p = 16$. **C** Paired differences against $p = 8$, as a percentage of the error at $p = 8$: on the legacy design (grey) every interval spans zero; on the smooth design (v2, black) every larger basis is better. Per-pair intervals; the $p = 128$ comparison also survives the selection adjustment (A29). DeepOKAN, a DeepONet with Kolmogorov–Arnold layers; FNO, Fourier neural operator.

5.02×10^{-4}, a factor of 28.7. The best DeepONet found anywhere, at the smallest basis, sits 53× above the floor of the largest. The edge rule is the rule this rung was most exposed on: at the two smallest bases DeepONet's best learning rate first sat on the top of the grid, still moving. The grid was extended one rate upward for those arms, the optimum stayed where it was and is now interior, and every selected rate in this rung is bracketed, so the Fourier operator of L5 is not compared with an under-tuned DeepONet. *That DeepONet is as accurate as an oracle handed the true coefficients of its first 4 modes.*

S5 The scope of the L6 null

L6 predicted that the benefit of a separate kinetic object grows as the Damköhler number falls, and the main text finds no support for it. Kinetic control means Damköhler well below unity, and this dataset never goes there: its lowest per-material value is 7.7 on the run horizon ($k_{\mathrm{LDF}}\, t_{\mathrm{final}}$, the axis of the L6 test) and 2.4 on the stoichiometric time. The hypothesis predicted that separating the kinetic object pays most where kinetics dominate, and we have found no support for it over a range whose slow-kinetics end is merely *less fast*, not slow. A design that sampled $\mathrm{Da} < 1$ could in principle recover the interaction. We did not sample it, and the reason is structural rather than an oversight: the kinetic coefficient is not a free parameter in this dataset but is derived from particle size through the Glueckauf relation, so $\mathrm{Da} \propto d_p^{-2}$ and reaching $\mathrm{Da} = 1$ from our slowest bed would take a pellet of about 9.5 mm. That is 11 pellet diameters along the 10 cm column, where a one-dimensional plug-flow model is approaching the limit of its validity, the regime Section 2.1 raises against our own, shallower anchor bed. Decoupling k_{LDF} from d_p to reach that regime would restore precisely the physical inconsistency this dataset was built to remove. The null therefore holds across the range a physically consistent bed of this kind can occupy, and is not a statement about the kinetically-controlled limit.

S6 The power of the L6 test

Why the test has power now, and its limit

The between-material standard deviation of the arm difference fell to 0.0104, or 18.5 % of the base error, from 69 % previously, a 3.8× noise reduction from Sobol sampling and physically consistent kinetics. With 240 clusters instead of 12, that is what turned an unanswerable question into an answerable one. Power is 82 % at a 4 % effect and 26 % at 2 %, with a size at the null of 1.8 % in this rung's own power simulation; the minimum detectable effect at 80 % power is 4.0 %. The observed effect sits just under it, so the point estimate may be optimistic and the lower bound is the conservative reading.

S7 The mode count and the n-width

The mode count is scoped to a sample size

An earlier version of the paper stated the mode count as a property of the map. It is not: it is what a given number of training materials buys. Measured against material count (Fig. 4D), modes clearing $R^2 > 0.2$ grow from 2 to 13 over 12 to 192 materials and are still climbing. Three counts of modes clearing $R^2 > 0.2$ appear in this paper, and they are the same statistic under different conditions: 9 on the legacy design, and 13 and 17 on the smooth design (v2) at the same number of training materials, the first for the learning-curve learner on its own basis and the second for the decomposition's regressor on the largest basis. They agree on what matters here: a learner predicts a modest set of mostly leading modes and little beyond it. A21 withdrew the earlier claim; the text that had replaced an earlier retraction repeated the same inference.

The n-width is real and was not what binds

The residual *energy* of this snapshot family decays algebraically, as $k^{-2.45}$ in the number of modes k, with $R^2 = 0.9910$ over $k = 8$–64. The residual *norm*, the square root of that and an average-case proxy for the Kolmogorov n-width (a worst case over all subspaces of that size), decays as $k^{-1.23}$ and halving the error requires 1.76× the modes. We separate the two because an earlier draft quoted the energy exponent and called it the n-width, next to a halving factor computed correctly from the norm: two numbers in one sentence that disagreed by a factor of two in the exponent. Any method whose output is a linear combination of a fixed basis is bounded below by it. Our frozen protocol called this the central result of the ladder and predicted that DeepONet would track it. The prediction was wrong: the bound is real, and it is inactive. Withdrawing it is A17. For transport-dominated problems the slow decay is established: Ohlberger and Rave [11] prove a *lower* bound of $\frac{1}{2}k^{-1/2}$ for linear advection with jump discontinuities, which is a bound and not a rate. Our measured decay is faster, which is consistent with it: the theorem covers fields with jump discontinuities, while ours are smoothed by axial dispersion and by finite-rate kinetics, so the worst case it constructs does not occur here. That is also why the bound can be real and inactive at the same time.

S8 Monotonicity of the predicted fronts

The two-front frame of Section S14 predicts two front trajectories per sample. The obvious first attack on its result is that the front predictor does not know the constraints the true trajectories satisfy exactly. It does violate them, in 55 % of samples at the lower landmark, while the true trajectories violate them in none. But projecting onto the constraint set changes R^2 by at most 1.9×10^{-4} and makes the reconstruction significantly *worse*: 0.04770 against 0.04812, a difference of -0.00041 with interval $[-0.00070, -0.00010]$ (tighter than Table S6's because the projected arm is a deterministic transformation of the raw arm, so the between-material variance cancels). The violations are frequent but negligible; the remaining headroom is the front predictor's raw accuracy.

S9 The dispersion closure, and the limits of the isotherm and the parameter space

Our datasets use $D_L = 0.7D_m + 0.5d_pu$, with $u = v/\varepsilon_t$ the interstitial velocity and D_m the molecular diffusivity of water in air, a form due to Wakao and Funazkri [12] that Perry's handbook [13] presents as a *lower bound* for *nonporous* particles; for porous, strongly adsorbing particles such as ours the effective axial dispersion is larger. The form is also the high-particle-Péclet *limit* of Edwards and Richardson's correlation [14], $D_L = 0.73D_m + 0.5\,d_pu\,[1 + 9.7\,D_m/(d_pu)]^{-1}$, whose mechanical term carries the extra bracketed factor, so we measured whether the limit applies. The particle Péclet number d_pu/D_m runs 4.6–190 across the dataset, median 49, spanning 1.62 decades. Over that range the two published forms disagree by a median factor of 1.19, exceeding 1.1× for 86.3 % of samples and 1.5× for 9.9 %, with a worst case of 2.04. The disagreement is bounded and modest for the dataset, but it is not modest for the anchor.

The anchor bed sits at a particle Péclet number of 2.9, and the disagreement between the two forms is maximised at 3.8, where it reaches 2.06×. In other words the one bed in this paper that is compared against an experiment is also the one place where the truncation matters most: there our closure disperses 2.02× more than the correlation it is the limit of, taking the column Péclet number from 2.32 to 1.15. Re-solving that bed under the untruncated form (a labelled post-hoc sensitivity, not a fit, and not the pre-registered result) moves the 95 % arrival time from 431 to 384 minutes against 354 measured, the plateau from 0.336 to 0.293 against 0.267, and the error against the digitised effluent from 0.128 to 0.105 (Figure S3). The authors' own

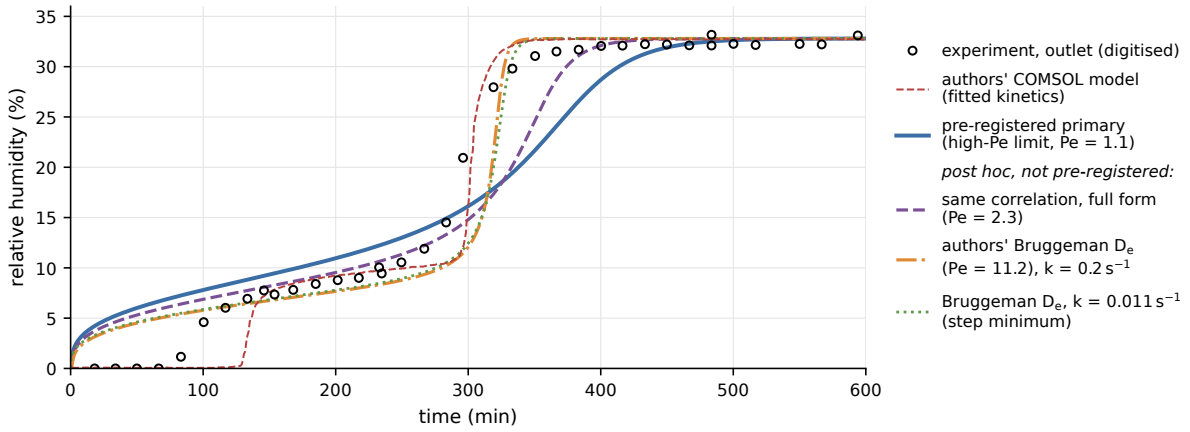


Figure S3: Post-hoc dispersion sensitivity on the anchor bed, labelled as such and reported as a limitation rather than as a result. The pre-registered primary uses the high-Péclet limit of the correlation; the second curve is the *same* correlation without that truncation, at the same particle size, velocity and molecular diffusivity; the third is the authors' own Bruggeman closure for the effective dispersion D_e; the fourth repeats that closure at the smallest LDF constant k (in s^{-1}) that still keeps the step. Pe is the column Péclet number. The measured 95 % arrival time is bracketed by the untruncated correlation and the authors' Bruggeman closure.

Bruggeman closure goes further, to 0.083, but overshoots in the other direction at 329 minutes; the measured value is bracketed by the two closures. A substantial part of the anchor's second named discrepancy is therefore one term evaluated at its limit: the untruncated form removes 61 % of the model's excess 95 % arrival time and 18 % of its error against the digitised effluent (the error is dominated by the first-wave timing, which no dispersion closure touches). None of this affects a surrogate result, because every arm sees the same reference; it bounds how far the reference may be read as physical.

The isotherm's low-humidity branch is too weak in the smooth design

In the anchor's MOF-303 parameterisation the Henry fraction and the step exponent were fitted to the published step position and capacity, never to dynamics, and the primary sites carry the smooth design's fixed, weak affinity ($b_H = b_C/20$); the anchor comparison shows that this isotherm holds too little water below the step. The MOF-like design samples the affinity instead, up to sites stronger than the step sites (main text, Section 2.1). Table S2 lists every run of the exploratory grid behind that statement (`anchor_strong_sites.py`; not pre-registered). Only the primary-site affinity and the Henry fraction change; nothing is fitted. The main text plots the run closest to the measured curve.

Part of the smooth design's heat range is occupied by no placed framework

The smooth design's sampled heat of adsorption runs from −60 to −38 kJ mol^{-1}; the water-harvesting frameworks we placed (Figure S1) report −61 to −49. Near its least exothermic edge the sampled range gives an isosteric heat ($-\Delta H + RT$) below water's latent heat of vaporisation at room temperature, so that water is bound more weakly than in its own liquid. That is the regime of hydrophobic adsorbents rather than of the frameworks this paper is motivated by. The range was pre-registered and is kept; verdicts averaged over it weight that region as heavily as the occupied one.

Not every sample reaches the step

A harvester is operated above its step by design; the sampled space is not. 14 % of the accepted samples of the smooth (v2) dataset (18 % of the legacy one) are fed below their material's step humidity, so the feed never reaches pore filling and the breakthrough is the primary-site

Table S2: Exploratory anchor grid (not pre-registered): the pre-registered primary configuration re-solved with the primary-site affinity $\log_{10}(b_H/b_C)$ at the smooth design's value, the MOF-like design's median and a strong site, and the Henry fraction f_H at the cited MOF-303 form and at the below-step fraction its source reports. Times are minutes to 5, 50 and 95 % of the inlet; "plateau" is the mean outlet fraction from 150 to 250 min; nRMSE is against the digitised effluent. The authors' COMSOL line has an nRMSE of 0.069.

$\log_{10}(b_H/b_C)$	f_H	t_5	t_{50}	t_{95}	plateau	nRMSE
−1.3	0.06	2	303	431	0.336	0.128
−1.3	0.17	3	270	388	0.366	0.100
+0.1	0.06	14	312	444	0.326	0.136
+0.1	0.17	34	297	423	0.331	0.105
+1.0	0.06	37	313	440	0.324	0.130
+1.0	0.17	84	300	414	0.319	0.094
measured		86	287	354	0.267	—

front alone. These samples are in every verdict; a surrogate built only for harvesting conditions would not need them.

The step does not move between feed temperatures

The cluster affinity is set per run so that the step sits at the material's step humidity at the feed temperature (Section 2.1). Within a run the step still moves with the local bed temperature through the heat of adsorption, but across runs a material's step humidity at the feed temperature is the same by construction. In real MOF-303 it moves with temperature [15]. The datasets therefore do not test whether a surrogate transfers the temperature dependence of the step.

Three parameters are not measurements

Pellet density and bed porosity are packing values, not MOF-303 measurements [16]. No citable measured heat capacity for MOF-303 was found, so it is reported as a sensitivity band (the grey thermal variants of Figure 2 of the main text) rather than as a cited constant.

No covalent organic framework case

We found no published breakthrough measurement on a covalent organic framework (COF) suitable for this comparison, so no claim is made for that class.

S10 When each pre-registration was fixed

Each pre-registration is a file in the public repository, fixed by the commit that added it. Table S3 gives, for each, that commit and the first commit of the results it governs, by author date. The repository's history was rewritten once, on 5 October 2026, before it was made public, to remove third-party files; the rewrite kept every author date and changed every commit identifier, and `COMMIT_MAP.tsv` maps each old identifier to its new one. The ordering below is therefore attested by the author's own history, not by an external registry. The original ladder protocol is the weakest case: it was written before its runs but committed only when the repository was created, minutes before the first results files, so for the rungs it governs (L3, the first L5 sweep and the coordinate change, all on the exploratory 12-material design) the order rests on the author's word. The two pre-registrations added for this submission were pushed to the public repository before anything was fitted or scored, so the push is an external timestamp; data generation for the MOF-like design began seven minutes before its push.

Table S3: Pre-registrations and the first results they govern (author dates, UTC+3). Commit identifiers are those after the history rewrite of 5 October 2026 (`COMMIT_MAP.tsv` maps them to the originals); for the rows fixed before that date the order rests on the author dates, for the later rows also on the public push.

pre-registration	governs	commit, fixed	first results
`03_LADDER_PROTOCOL.md`	L3, first L5 sweep, coordinate change	81295d4, 30 Aug 2026, 17:50	30 Aug 2026, 17:57
`PREREG_L6_v2.md`	L6	02c6cf1, 1 Sep 2026	2 Sep 2026
`PREREG_L1L2L7_v2.md`	L1, L2, L7	db7093b, 2 Sep 2026	3 Sep 2026 (L1, L7), 5 Sep (L2)
`PREREG_L4b_v2.md`	L4	4389c07, 2 Sep 2026	24 Sep 2026; corrected re-run 11 Oct
`PREREG_LASSITTER.md`	the MOF-303 anchor	5dfdf50, 3 Sep 2026, 09:56	3 Sep 2026, 10:03
`PREREG_L5_v2.md`	L5 on 240 materials	4ecc5d9, 28 Sep 2026	4 Oct 2026
`PREREG_BASELINES.md`	Gaussian process, coarse solve, fitted closed forms	073ad4d, 4 Oct 2026	11 Oct 2026
`PREREG_DD_ADDITIONS.md`	screening, named frameworks	071ca9a, 25 Oct 2026 (pushed)	25 Oct 2026, after the push
`PREREG_DD_V3.md`	MOF-like design, DeepONet's own floor	24e72e6, 25 Oct 2026, 13:23 (pushed 13:34)	25 Oct 2026, 18:22 (dataset generated from 13:27, nothing fitted or scored before 18:22)
amendment 1	converged floor, controls, t90	e1433b3, 26 Oct 2026, 02:16 (after the first v3 results)	26 Oct 2026, after the push

S11 Fixed properties, numerics and the Damköhler number

Fixed properties

Every run uses $L = 0.10\,\mathrm{m}$, $D_{\mathrm{in}} = 0.05\,\mathrm{m}$, $\rho_g = 1.2\,\mathrm{kg\,m^{-3}}$, $C_{pg} = 1000$ and $C_{ps} = 1000\,\mathrm{J\,kg^{-1}\,K^{-1}}$, $k_z = 0.10\,\mathrm{W\,m^{-1}\,K^{-1}}$ and $h_w = 10\,\mathrm{W\,m^{-2}\,K^{-1}}$. Axial dispersion follows the packed-bed correlation $D_L = 0.7D_m + 0.5\,d_p v/\varepsilon_t$ with $D_m = 2.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$. In the smooth (v2) dataset the kinetic coefficient is not sampled but derived from the particle by the Glueckauf relation,

$$k_{\mathrm{LDF}} = \frac{15\,\varepsilon_p D_m/\tau_p}{(d_p/2)^2\,\rho_p\,K}, \qquad K = \frac{q^*(c_{\mathrm{in}}, T_{\mathrm{in}})}{c_{\mathrm{in}}}, \tag{S1}$$

with pellet porosity $\varepsilon_p = 0.35$, tortuosity $\tau_p = 3$ and $D_m = 2.6 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$. The generator therefore uses two slightly different values of the molecular diffusivity, one in each correlation; we report both rather than reconcile them after the data were generated.

Numerics and storage

Equations (1)–(3) are discretised by the method of lines on 2000 cell-centred finite volumes (first-order upwind advection, central dispersion and conduction) and integrated with SciPy's implicit backward-differentiation-formula (BDF) method at relative and absolute tolerances 10^{-6} and 10^{-10} to 600 output times; Section 2.1 verifies the solver. Each run is integrated to 2.5 stoichiometric times; if the outlet has not reached the acceptance level the horizon is extended to 5, 12.5 and 30 stoichiometric times in turn, which 52 % of the accepted runs of the smooth (v2) dataset needed. A draw is rejected before solving if its feed loads too little or too much of capacity, its Henry constant is not finite and positive, its isothermal mass-transfer zone is not resolved by the grid, or its stoichiometric time is implausible; 133 of the smooth (v2) dataset's draws were rejected and 3947 accepted. Fields are stored as c/c_{in}, $q/q_{\max}$ and T/T_{in} on a uniform grid in z/L and t/t_{final}, at 256×256 for the smooth (v2) dataset and 512×256 for the legacy one.

Damköhler number

The Damköhler number used throughout is computed per sample as $\mathrm{Da} = k_{\mathrm{LDF}}\,t_{\mathrm{final}}$, and rung L6 regresses on its per-material median. Because the horizon is a multiple of the stoichiometric time that depends on whether it was extended, Da carries that multiple as well as the kinetic ratio $k_{\mathrm{LDF}}t_{\mathrm{st}}$; for the 52 % of runs that were extended it is larger than the kinetic ratio by the extension factor. We keep the definition the pre-registration used and state this property rather than redefine the axis after the result.

S12 The parameter space

Table S4 lists the sampled ranges of both designs; the legacy design of the exploratory comparisons sampled the kinetic constant directly instead of deriving it from the pellet size (Section S5).

S13 The surrogates of each rung

Table S5 gives the architecture, training and selection of every surrogate of the seven explanations. The analyses added for this submission use the settings of the rungs they extend: POD with gradient boosting and DeepONet at $p = 16$ as in L5, the coarse solve of the baselines pre-registration, and the floor solver of the main text (Section 3.2).

Table S4: The parameter spaces of the smooth (v2) and MOF-like (v3) designs. A "material" here is the upper block: an idealised framework pellet described by its isotherm (capacity, step position, step steepness, the fraction of capacity on the low-humidity sites and, in v3, their affinity relative to the step sites, heat) and by pellet size, density and packing. Capacity 8–30 $\mathrm{mol\,kg^{-1}}$ is 0.14–0.54 g of water per g. The step is re-anchored at each feed temperature so that it sits at the stated relative humidity, so one material's isotherm is not the same curve at every condition. All parameters are sampled by scrambled Sobol and mapped linearly; both designs use the same operating conditions. Every surrogate except rung L6 receives exactly these values, standardised on its own fitting set, and L6 receives observable descriptors instead.

parameter	symbol	units	smooth (v2)	MOF-like (v3)
material				
saturation capacity	$q_{\max}$	$\mathrm{mol\,kg^{-1}}$	8–30	8–56
heat of adsorption	ΔH	$\mathrm{kJ\,mol^{-1}}$	−60 to −38	−62 to −40
step position (relative humidity)	$\mathrm{RH_{step}}$	–	0.08–0.45	0.05–0.45
cooperativity exponent	n	–	1–6	1–32
Henry-site fraction	f_H	–	0.03–0.25	0.03–0.3
primary-site affinity	$\log_{10}(b_H/b_C)$	–	fixed at $-\log_{10} 20$	−1.3 to 1.5
particle diameter	d_p	mm	1.5–5	1.5–5
particle density	ρ_p	$\mathrm{kg\,m^{-3}}$	700–1400	700–1400
bed porosity	ε_t	–	0.3–0.48	0.3–0.48
operating condition (both designs)				
feed relative humidity	$\mathrm{RH_{feed}}$	–	0.15–0.85	
superficial velocity	v	$\mathrm{m\,s^{-1}}$	0.03–0.3	
feed temperature	T_{in}	K	288.15–313.15	

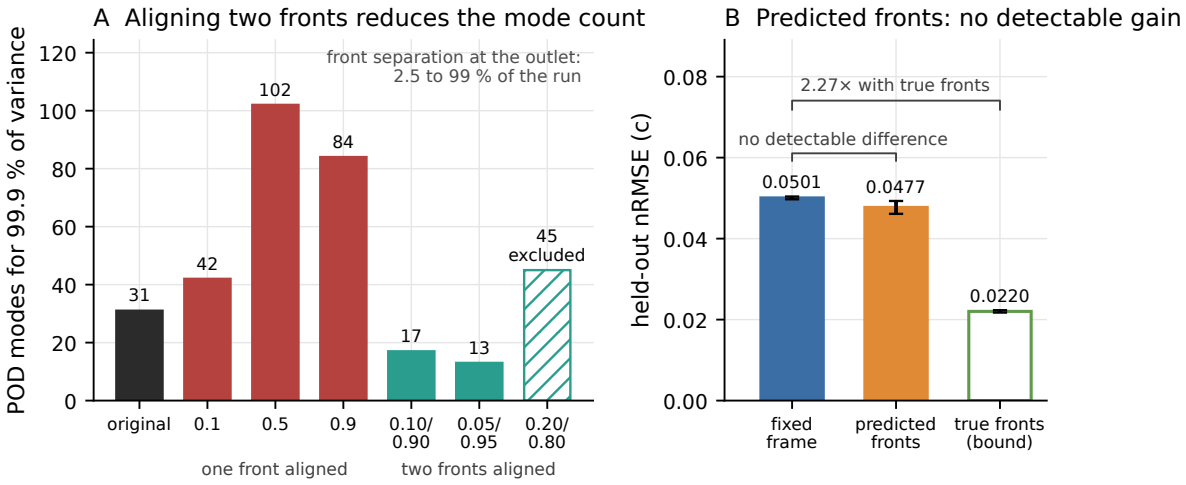


Figure S4: The two-front frame (legacy design). **A** Modes needed for 99.9 % of training variance in each frame: aligning one front raises the count, aligning both lowers it. Tick labels give the level of c/c_{in} at which each aligned front is located (one level, or a pair); the hatched pair was excluded by its interpolation guard. **B** Held-out error: with predicted fronts the two-front frame is not distinguishable from the fixed frame (the interval runs from 24 % better to 16 % worse, 12 materials); handed the true fronts (hollow) it is better, an upper bound rather than a result.

Table S5: Surrogates by rung. Absolute errors in different rows are not comparable. POD, proper orthogonal decomposition; NTK, neural tangent kernel; L-BFGS, limited-memory Broyden–Fletcher–Goldfarb–Shanno; DeepOKAN, a DeepONet with Kolmogorov–Arnold layers; p, basis size.

rung	data, design	input → output; encoding	architecture	training
L1	smooth (v2), five-fold	parameters → POD coefficients per channel	ridge; random forest; gradient boosting; perceptron with standardised targets	scikit-learn; perceptron Adam with early stopping
L1 curve	smooth (v2), five-fold	parameters → 32 POD coefficients of c; training materials subsampled	one gradient-boosted regressor per mode (not the L1 perceptron)	three seeds per size
L2	smooth (v2), five-fold	as L1	perceptron width and depth, boosting depth, forest leaf size, swept in both directions	as L1
L3	legacy	parameters → POD coefficients; full stored field	perceptron and two Kolmogorov–Arnold [17] families (Gaussian radial basis, Chebyshev), depth 3, matched to 50k/200k/800k parameters; widths perceptron 114/269/583; RBF 45/98/203; Chebyshev 48/103/215	Adam, cosine schedule, 12,000 steps, per-family learning-rate grid
L4	smooth (v2); time axis and single split	(parameters, z, t) → (c, q, T) pointwise, bounded output head	one perceptron, 4 layers of 192 (114,435 parameters); data-only twin and physics-weighted arms (fixed, gradient-norm, NTK, self-adaptive); residual of equations (1)–(3) with the Danckwerts inlet conditions, corrected (B72)	Adam, cosine schedule, 8000 steps of 4096 data, 1024 collocation and 512 boundary points; then L-BFGS (polish)
L5	legacy	parameters → c field, subsampled from the stored field	DeepONet and DeepOKAN over basis size p; Fourier neural operator over mode count; all matched to 200k parameters	Adam, cosine schedule, 8000 steps, per-arm learning-rate grid
L5 basis size	smooth (v2), five-fold	as L5	DeepONet over basis size p, matched parameters	as L5, step budget tripled
L5 basis vs. map	legacy; smooth (v2), five-fold	parameters → POD coefficients of c over basis size	one gradient-boosted regressor per mode	one seed (legacy), three seeds (smooth v2)
L6	smooth (v2), five-fold	54 observable descriptors (isotherm at three temperatures, bed and operating values), z, t → (c, q, T)	joint perceptron versus separate equilibrium, kinetic and transport heads, with q from equation (2); both 120k parameters, depth 4	Adam, cosine schedule, 12,000 steps
L7	smooth (v2), five-fold, paired with L1	parameters → outlet curve	equilibrium shock, Klinkenberg, constant pattern; nothing fitted	none
Warp	legacy	parameters → c, in a fixed frame or the two-front frame of equation (S2)	POD plus one gradient-boosted regressor per mode; front trajectories from a POD of the landmark curves	three seeds; oracle arm given the true fronts

Table S6: The two-front frame against the fixed frame (legacy design). Differences are the fixed frame's nRMSE minus the arm's, so positive means the arm is better; CI is the cluster-bootstrap confidence interval. The oracle arm is handed the true front trajectories and is *not* comparable with any arm that is not; it sizes the remaining headroom and is not a result.

arm	held-out nRMSE	verdict vs. the fixed frame
fixed frame	0.05005	reference
two-front, predicted fronts	0.04770	0.00235, CI [−0.00812, 0.01183]: no difference
two-front, oracle fronts	0.02203	0.02802, CI [0.01990, 0.03668]: significant

S14 The coordinate change, and where the obstruction moved

S14.1 Scope and prior art

Aligning multiple transports is not new, and none of it is claimed here. The two-transport case is the founding example of the shifted POD [18], multi-front variants follow [19–23], a neural-network shifted decomposition predates the last of these [24], and breakthrough curves have been decomposed with a single characteristic time [25]. Our two-landmark linear warp is piecewise landmark registration from functional data analysis [26] (a landmark-free alternative is in [27]); Zucatti and Zahr [28] apply the same primitive to shock features in reduced-order models of convection-dominated problems, and registration-based reduction is an established family [29, 30]. What is ours is the measurement.

S14.2 One shift makes it worse; two make it much better

The standard remedy for a transport-dominated n-width is to factor the front position out before projecting [18]. We warped each field onto its own local arrival time and swept the alignment level. *It makes the n-width worse*: aligning on the half-height crossing takes the modes needed for 99.9 % of training variance from 31 to 102. It does not help accuracy even given the exact front trajectory for free: handed the oracle warp, the co-moving frame still loses to the fixed frame. The reason was measured before it was invoked: the separation between the two waves spans a factor of 40, and one shift cannot align two waves whose separation varies that much.

Using *two* shifts, one per wave, and normalising the interval between them (Figure S4),

$$\sigma(z,t) = \frac{t - t_{\mathrm{lo}}(z)}{t_{\mathrm{hi}}(z) - t_{\mathrm{lo}}(z)}, \tag{S2}$$

collapses the n-width where one shift inflated it: 31 → 17 modes, or 13 for the wider landmark pair. A third pair was excluded because a narrow window divides by a small span and its interpolation floor reached 29 % of the fixed-frame error it is judged against, above the guard's threshold of 10 %.

S14.3 The verdict, what it can resolve, and the relocation

Table S6 gives the result: three seeds for every stochastic component, both arms computed in the same run, and a paired cluster bootstrap by material.

The predicted-front arm does not beat the fixed frame. Given the true front trajectories, the same POD-plus-regressor arm is 2.27× better than the fixed frame, which is what perfect front location would buy, and 2.17× better than with predicted fronts. On 12 held-out materials the predicted-front interval admits anything from a 24 % improvement to a 16 % degradation; it rules out the oracle's gain, not a modest one. With that scope, this is a relocation of the obstruction. Through L1–L5 the obstruction was the parameter-to-coefficient map, at the number of modes a given training set supports. It is now "two smooth monotone one-dimensional curves are not

Table S7: The accuracy–cost frontier, in solve-equivalents. Each row is a point on the learning curve of Section 3.1, priced; its error is that of the curve's per-mode gradient-boosting learner. The last column is a *lower* bound on the number of screening queries at which a surrogate becomes cheaper than the production solve; it needs only the data cost, so it holds for any learner and whatever a query costs. The main text (Section 3.3) prices DeepONet's fit and compares against the coarse solve, which moves the break-even point to about 30,000 queries.

training materials	training simulations	held-out nRMSE	break-even queries $\geq$
12	196	0.0654	198
24	395	0.0531	397
48	790	0.0465	791
96	1577	0.0402	1578
192	3158	0.0348	3159

located accurately enough", a better-posed problem, with a target a front-locating model can be held to. The oracle's gain is not unique to this route: the Fourier operator of L5 reaches a comparable error on the same held-out materials without any oracle.

Monotonicity is not the lever

The predicted fronts violate the constraints the true ones satisfy in 55 % of samples, but projecting them onto the constraint set makes the reconstruction slightly worse (Section S8); the remaining headroom is the front predictor's raw accuracy.

Relation to constant-pattern behaviour

Equation (S2) is not constant-pattern scaling [31]: when the two fronts move at different rates and their separation varies 40-fold, no single scale factor aligns both, which is why the second landmark does the work.

S15 The accuracy–cost frontier

The learning curve is a price list: going from 12 to 192 training materials multiplies the data cost by 16× and divides the error by 1.88×. The rows are priced for the curve's per-mode regressor; the L1 perceptron is already more accurate, and L2 found a configuration 21.3 % better still, so a better learner shifts the whole frontier down. The exchange rate between data and error is the transferable part.

The eliminated rungs are interventions that add cost without buying accuracy: a Kolmogorov–Arnold map costs fit time and nothing else, and the closed forms are cheaper but worse even when fitted. Of the levers that remain, only training materials cost simulations; a better configuration, a small residual weight (9 %) or a different operator cost fit time or collocation points. Every statement here is cost *at a stated accuracy*: the production solve answers exactly, the surrogate with the held-out error in the third column, and the coarse solve of the main text with 0.62 % at a fraction of the production cost.

S16 Corrections and retractions

30 claims made in this project were withdrawn and 93 further defects were caught and corrected; Supplementary S1 lists every entry, and the text cites them by code (A for a withdrawn claim, B for a defect). We report them because six reversed a headline verdict, and a reader who does not know which numbers moved cannot evaluate the ones that did not; disclosed self-correction is also what lets a reader judge the rest [32].

Six that reversed a headline

Two reversed the physics-as-a-loss rung: its legacy claim that the residual hurts did not survive a sweep of weightings (B69), and every one of its verdicts was then withdrawn when the residual proved to be the wrong equation (B72, A28), until the corrected re-run reported the small benefit of rung L4 (Section 3.1). A17 withdrew the explanatory role of the Kolmogorov n-width; A22 turned a null described as decisive into one that excludes only very large effects (A25 later corrected its power simulation in the opposite direction); A29 retracted the flat-in-basis verdict of L5 on its 240-material re-measurement; and A30 revised the cost case once the coarse-solver baseline proved more accurate than the perceptron surrogates. B75, which removed "significantly" from the Fourier operator's advantage over DeepONet once the choice of both arms was paid for, is not counted: the direction of that comparison stands.

The errors recur in five classes: a property measured at one sample size and stated as intrinsic (A18, A21, A26, A29); normalising by a quantity that can vanish (A15, B34, B41); comparison against a weak configuration of the competitor (A20, B14, B22, B23); undertrained arms scored as architectures (A19, B24); and power and calibration (A13, A22, A25).

S17 The seven pre-registered explanations in full

Moved verbatim from the main text (Section 3.1), which summarises them. Every explanation was tested on the smooth design (v2) or, where marked exploratory, on the legacy design.

Figure 3 of the main text summarises the seven explanations in the verdict words fixed before each run; the headings below paraphrase them, and the pre-registrations give them in full.

L1, more training materials: not eliminated

The best fixed-basis surrogate, a multilayer perceptron on 64 POD coefficients per channel, reaches 0.0306 and beats gradient boosting in all five folds. On a learning curve over training materials, held-out error falls by 1.88× from 12 to 192 materials and still improves significantly at the last step, as $N_{\mathrm{mat}}^{-0.222}$ (exponent interval [0.205, 0.239]). Refitting the basis on each subset changes nothing (0.0348 against 0.0348 at 192 materials): none of the gain is in the basis. A Gaussian process on the same coefficients finds no difference between the perceptron and the Gaussian process (0.0309 against 0.0306; the smallest difference the test could detect is 6 %).

L2, more capacity: not eliminated

Of 28 configurations swept in both directions, 3 sweeps have interior optima, but the random forest is still improving at its fully grown limit, so the pre-registered rule keeps capacity open. The best configuration, the perceptron of depth 8, is 21.3 % better than the L1 setting, 17.8–24.8 % once its selection is paid for.

L3, Kolmogorov–Arnold maps: eliminated (exploratory)

At matched parameter counts, the perceptron beats both Kolmogorov–Arnold [17] families at every budget, 4 of 6 comparisons surviving the selection adjustment and none reversed, on the 12-material design. The literature is split on this comparison [33–35]; our statement is narrow.

L4, the governing equations as a training loss: helps at small weights

A fixed residual weight of 10^{-5} improves transfer to 48 held-out materials by 9 % over an identical data-only network (interval over every weighting tried: [0.00119, 0.00285]); every larger weight and every adaptive weighting scheme is worse, and minimising the residual at inference makes predictions about 3 times worse. Physics-informed surrogates of adsorption processes gain most when data are scarce [36]; with the training data used here the gain is small, and the result rests on a single split.

L5, a better operator: a small gain from the basis, complete by 16 modes

DeepONet improves by 12 % from $p = 8$ to $p = 128$ on the 240-material design, interval [0.00088, 0.00173] after paying for the learning-rate choice, and the whole gain is reached by $p = 16$. A Fourier neural operator [37] is more accurate by point estimate on the 12-material design, but not significantly once its selection is paid for (exploratory).

L6, separating kinetics from equilibrium: a real benefit, not the predicted one

Separate equilibrium, kinetic and transport heads beat a joint network by 0.00210 (interval [0.00056, 0.00363]), in all folds and seeds, but the benefit shows none of the dependence on Damköhler number that the hypothesis predicted; the dataset never reaches kinetic control, so that null is scoped to near-equilibrium beds.

L7, closed forms: eliminated

Unfitted, the best classical approximation (Klinkenberg's [38, 39]) is 3.73× worse than learning on the outlet curve; with two physically interpretable corrections fitted on the training materials it is still 2.44× worse.

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